Int J Life Cycle Assess
https://doi.org/10.1007/s11367-017-1405-9

LCA FOR MACHINES

Life cycle assessment of copper production: a case study in China

Jinglan Hong[1] · **Yilu Chen**[2] · **Juan Liu**[3] · **Xiaotian Ma**[1] · **Congcong Qi**[1] · **Liping Ye**[1]

Received: 8 June 2016 / Accepted: 9 October 2017
© Springer-Verlag GmbH Germany 2017

Abstract
Purpose China is the world's largest producer and consumer of refined and reclaimed copper because of the rapid economic and industrial development of this country. However, only a few studies have analyzed the environmental impact of China's copper industry. The current study analyzes the life cycle environmental impact of copper production in China.
Methods A life cycle impact assessment using the ReCiPe method was conducted to estimate the environmental impact of refined and reclaimed copper production in China. Uncertainty analysis was also performed based on the Monte-Carlo simulation.
Results and discussion The environmental impact of refined copper was higher than that of reclaimed copper in almost all categories except for human toxicity because of the direct atmospheric arsenic emission during the copper recycling stage. The overall environmental impact for the refined copper production was mainly attributed to metal depletion, freshwater ecotoxicity, marine ecotoxicity, and water depletion potential impact. By contrast, that for the reclaimed copper production was mainly caused by human toxicity impact.
Conclusions Results show that the reclaimed copper scenario had approximately 59 to 99% more environmental benefits than those of the refined copper scenario in most key categories except for human toxicity, in which a similar environmental burden was observed between both scenarios. The key factors that reduce the overall environmental impact for China's copper industry include decreasing direct heavy metal emissions in air and water, increasing the national recycling rate of copper, improving electricity consumption efficiency, replacing coal with clean energy sources for electricity production, and optimizing the efficiency of copper ore mining and consumption.

Keywords Environmental benefit · Life cycle assessment · Refined copper · Reclaimed copper · Regional level

1 Introduction

With the rapid economic and industrial development of China, the production of refined copper products consistently increased during the last decade (China Statistical Yearbook 2005–2015) from approximately 2.04 million tons in 2004 to 7.96 million tons in 2014, thereby indicating an average annual increase of 14.58%. According to statistical data, China is the world's largest producer of refined and reclaimed copper, which account for 31.48 and 58.12%, respectively, of the total global output in 2013 (China Statistical Yearbook 2014; Wang et al. 2015). However, China, which is also the world's largest consumer of copper, faces numerous challenges. The Chinese copper industry suffers from low electricity efficiency and copper recovery rate, as well as the overuse of raw materials (HJ 559-2010 2010). The percentage of

Jinglan Hong and Yilu Chen contributed equally to this work.

Responsible editor: Zuoren Nie

✉ Jinglan Hong
hongjing@sdu.edu.cn

[1] Shandong Provincial Key Laboratory of Water Pollution Control and Resource Reuse, School of Environmental Science and Engineering, Shandong University, Jinan 250100, People's Republic of China

[2] College of Biological and Pharmaceutical Engineering, Nanjing Tech University, Nanjing 210009, People's Republic of China

[3] China Environmental United Certification Center, No. 1 Yuhuinanlu, Chaoyang District, Bejing 100029, People's Republic of China

Published online: 23 October 2017

reclaimed copper of that in the total refined copper was merely 22.8% in China in 2013, thereby suggesting a significant potential for improvement (China Nonferrous Metals Industry Association 2015). Moreover, pollutants, such as particulates, SO_2, CO_2, NO_2, solid waste, and wastewater, which are generated from copper manufacturing processes, may seriously damage the environment and the health of the population of neighboring areas (Song et al. 2014; Zhao et al. 2008). Copper, which is a nonrenewable resource, is crucial in economic development. Therefore, systematically evaluating the environmental impacts of copper production in China is crucial to identify key factors in the industrial life cycle that may alleviate the shortage of copper resources, reduce the energy consumption, and decrease environmental pollution.

Life cycle assessment (LCA) is a beneficial tool to assess the environmental impacts of products, processes, or activities generated during their entire life cycles. The environmental impacts of the copper industry are extensively studied worldwide (Giurco et al. 2006; Song et al. 2014; Rubin et al. 2014; Memary et al. 2012; Reuter et al. 2015; Ayer et al. 2016). However, most previous studies only focused on several impact categories (Giurco et al. 2006; Memary et al. 2012; Rubin et al. 2014; Ayer et al. 2016) and failed to collect actual field data on direct water and air emissions. Moreover, studies (Song et al. 2014; Streicher-Porte and Althaus 2010; Li and Guan 2009) that have employed the LCA approach on copper production in China are limited in international peer-reviewed journals. Song et al. (2014) studied the environmental burdens in China from copper production. However, a percentage of the inventory data of this research were obtained from the GaBi database, thereby possibly leading to inaccurate results because of the differences in energy structures and technologies between China and Europe. Similarly, Streicher-Porte and Althaus (2010) reported a LCA study of copper production in China on the basis of reported Chinese industry data between 2006 and 2009 and ecoinvent database. No wastes such as slag, filter ashes, dusts, and sludge were involved. Li and Guan (2009) presented a conference article on the process of recycling copper based on the LCA method. However, their data sources and key factors were unclear (ISO 14040 2006a, b); both the energy input and pollution output information were inaccurate in their study. Furthermore, no study performed a sensitivity or uncertainty analysis. Thus, the results were unreliable. To solve these problems, an LCA of copper production in China was conducted. The goals of the current study were to quantify and compare the environmental impacts of refined and reclaimed copper production, identify their key factors, explore potential methods for environmental improvements, and determine the level of confidence in the assertion that reclaimed copper is more environment-friendly than refined copper.

2 Scope definition

2.1 Functional unit

The functional unit provides a quantified reference for all related inputs and outputs of a product. In this study, the production of 1 ton of copper (i.e., refined or reclaimed copper) was selected as the functional unit. All raw materials and energy consumption, road transport, direct emissions (i.e., air pollutants, wastewater emission, and solid waste generation), and waste disposal management were based on this functional unit.

2.2 System boundary

System boundaries were established by applying a cradle-to-gate approach for two scenarios, namely, the refined and reclaimed copper scenarios (Fig. 1). The former began with copper ore mining, whereas the latter started from old copper collection. Each of the processes considered raw materials and energy consumption, road transportation, direct pollutant emissions, and waste disposal. In particular, the industrial waste gangue generated from mining was dealt with the storage method, solid waste from copper refining and slag produced from copper reclaiming were reused as building materials, and waste refractory matter and anode were recycled. By contrast, other solid and hazardous wastes, including arsenic residues generated from copper refining or reclaiming, were disposed via sanitary landfills.

2.3 LCIA methodology

A life cycle impact assessment (LCIA) was conducted at the midpoint level by employing the ReCiPe method, which is one of the most commonly used indicator approaches in the LCA analysis (Goedkoop et al. 2009; Schryver et al. 2009). This method was built on the results of Eco-indicator 99 (Goedkoop and Spriensma 2001) and Centrum voor Milieukunde Leiden (i.e., CML 2000) (Guinée et al. 2001) with some modifications on LCA. The ReCiPe approach defines 18 midpoint categories: climate change, ozone depletion, human toxicity, photochemical oxidant formation, particulate matter formation, ionizing radiation, terrestrial acidification, freshwater eutrophication, marine eutrophication, terrestrial ecotoxicity, freshwater ecotoxicity, marine ecotoxicity, agricultural land occupation, urban land occupation, natural land transformation, water depletion, metal depletion, and fossil depletion. Particularly, potential environmental effects of future extractions are included in the LCIA results. In addition, normalization which is determined by the ratio of the impact per unit of emission divided by the per capita world impact for the year 2000 (Wegener Sleeswijk et al. 2008) was applied in this study to analyze the respective share of each midpoint

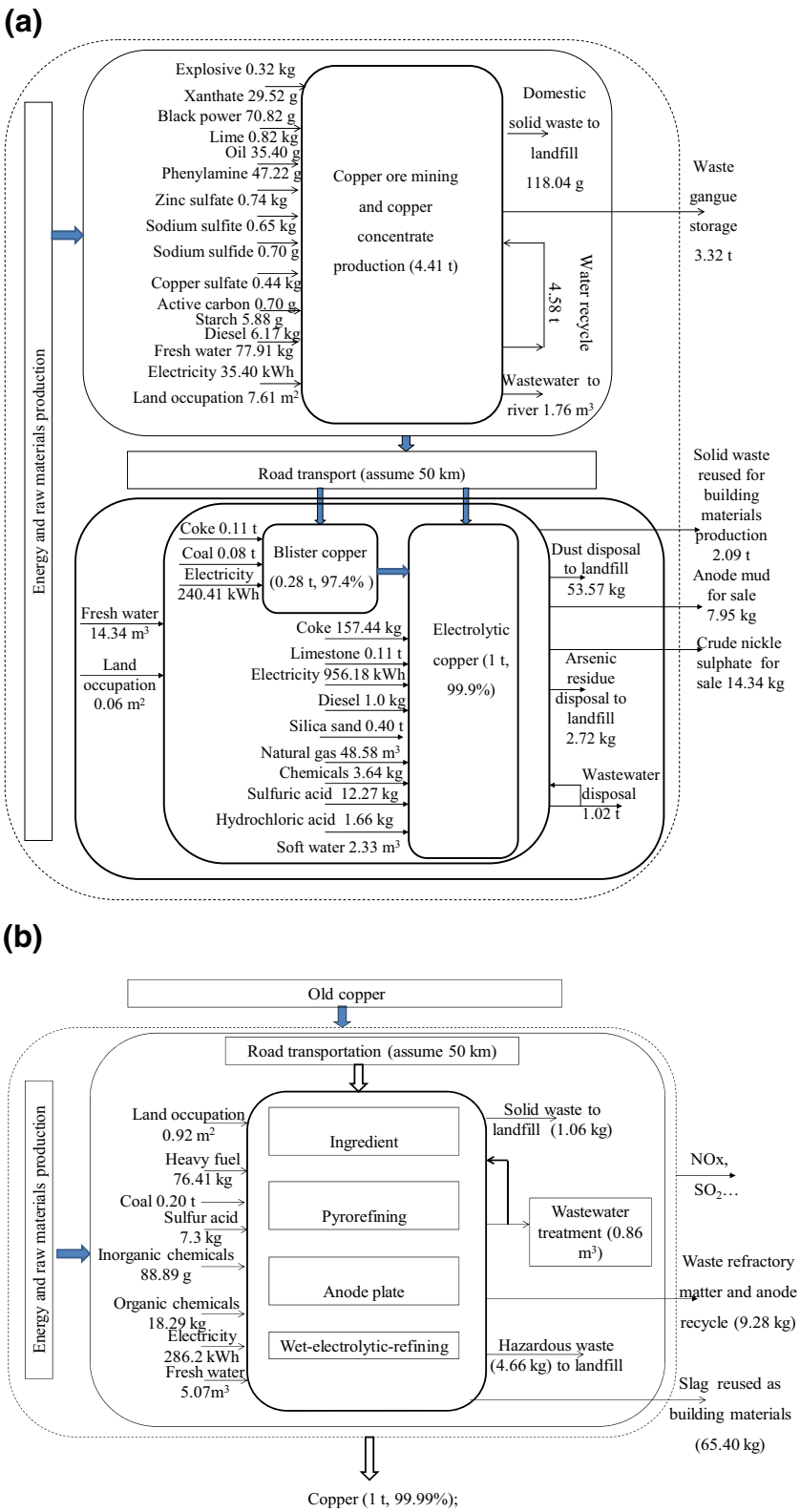


Fig. 1 System boundary and materials flow diagrams for the refined copper scenario (**a**) and the reclaimed copper scenario (**b**)

impact to the overall impact as well as to present comparable midpoint impacts. Detailed methodology and characterization factors of the ReCiPe model are available on the website of Institute of Environmental Science at Leiden University in the Netherlands (http://www.cml.leiden.edu/research/industrialecology/research projects/finished/recipe.html). Uncertainty analysis was conducted based on Monte-Carlo simulation.

2.4 Life cycle inventory and data sources

Table 1 shows the life cycle inventory (LCI). The LCI data presented in this study were collected from two refined and reclaimed copper production sites in China. Mass allocation regarding all the inputs and outputs of refined copper scenario is performed in this study. The annual capacities for refined copper using double side blowing bath smelting technology and reclaimed copper were approximately 130 and 35 kt, respectively, in 2014. The copper content rate of copper ore and copper concentrate is around 0.8 and 23.7%, respectively. The heavy metals (i.e., lead, copper, arsenic, and zinc) and sulfur content rates of refined copper scenario are shown in Table 2. The reviewed LCI in China for the national average coal-based electricity generation (Chen et al. 2016), road transportation (Chen et al. 2015), municipal solid waste disposal (Hong et al. 2017a), hazardous waste (Hong et al. 2017b), and wastewater treatment (Xu et al. 2015) were applied in the current study. The data on chemicals, natural gas, and heavy fuel oil production in Europe (Ecoinvent centre 2015) were adopted because of lack of information in China. To reduce the regionalized effect from Europe, the Chinese data on the aforementioned national coal-based electricity generation (Chen et al. 2016) and road transport (Chen et al. 2015) were used to adjust the European data to the Chinese situation. The LCA was conducted using the SimaPro 8.2 software.

3 Results and discussion

3.1 LCIA results

Table 3 compares the LCIA results for both refined and reclaimed copper scenarios. The environmental impact of refined copper was higher than that of reclaimed copper in almost all categories except for human toxicity and natural land transformation. In particular, the impact of refined copper was substantially higher (i.e., almost or over 10 times) on freshwater eutrophication, terrestrial ecotoxicity, freshwater ecotoxicity, agricultural land occupation, and metal depletion than that of the reclaimed copper industry. For ozone depletion and natural land transformation, the two scenarios had similar impacts.

Furthermore, the normalized LCIA midpoint scores for the entire life cycle in each scenario (Fig. 2) indicated that the environmental impact of the refined copper production industry was considerably higher than that of the reclaimed copper industry. In the refined copper scenario, metal depletion, freshwater ecotoxicity, marine ecotoxicity, and fossil depletion were the dominant contributors to the overall environmental impacts. Climate change, terrestrial acidification, freshwater and marine eutrophication, human toxicity, photochemical oxidant formation, particulate matter formation, and terrestrial ecotoxicity had minor to medium contributions, whereas the other impact categories could be disregarded. However, for the reclaimed copper scenario, only the environmental impact to human toxicity was significant. Climate change, terrestrial acidification, photochemical oxidant formation, particulate matter formation, fossil depletion, and freshwater and marine ecotoxicity had minor contributions and all the other impact categories could be disregarded.

To obtain an improved understanding of the contributions of the dominant pollutants to the aforementioned major impact categories, the contributions of the key substances to these categories were presented in Fig. 3. In both scenarios, the contribution of carbon dioxide to climate change was the highest, followed by methane. Mercury and arsenic emissions contributed most to human toxicity in the refined and reclaimed copper scenarios, respectively. The contribution of copper emission to water was the highest for freshwater and marine ecotoxicity in the refined copper industry, and the use of copper was the sole factor of metal depletion. For the reclaimed copper scenario, bromine, nickel, and copper emissions to water contributed the most to freshwater and marine ecotoxicity, whereas copper and arsenic emission to air played another dominant role in marine ecotoxicity. The dominant substances that cause metal depletion included copper, iron, and nickel. Compared with the reclaimed copper industry where coal and oil similarly contributed to fossil depletion, coal played a dominant role in fossil depletion in the refined copper scenario.

Figure 4 shows the key processes in the production of refined and reclaimed copper and their contributions to the 18 midpoint environmental categories. In both scenarios, direct emission and electricity were important contributors for categories such as climate change, terrestrial acidification, marine eutrophication, human toxicity, photochemical oxidant formation, particulate matter formation, and water depletion. Copper mine and crude copper used during the production of refined copper were the main contributors to freshwater ecotoxicity, marine ecotoxicity, agriculture land occupation, urban land occupation, and metal depletion. Heavy fuel oil consumption during the reclaimed copper production contributed to most categories except for climate change, marine eutrophication, and human toxicity. Furthermore, organic chemicals played an important role in the freshwater eutrophication, terrestrial ecotoxicity, ionizing radiation, agriculture land occupation, water depletion, metal depletion, and fossil depletion categories during the production of reclaimed copper.

3.2 Sensitivity analysis

Table 4 presents the sensitivity analysis results. For the refined copper scenario, the effect of the variation of electricity consumption was the highest on climate change and fossil depletion in which a 5% electricity reduction could decrease 40.57 kg CO_2 eq and 9.75 kg oil eq. The effect of the changes of direct emission and copper mines was the highest in the

Table 1 Life cycle inventories for the refined copper and reclaimed copper scenarios. Values were presented per functional unit

		Amount		Unit
		Refined copper	Reclaimed copper	
Resource and materials consumption	Land occupation	0.06	0.92	m^2a
	Water	14.34	5.07	t
	Electricity	1196.59	286.62	kWh
	Oil	–	76.41	kg
	Coal	0.08	201.15	kg
	Copper concentrate	4.41	–	t
	Limestone mine	0.11	–	t
	Coke	0.16	–	t
	Diesel	1.00	–	kg
	Silica sand	0.40	–	t
	Chemicals inorganic	3.64	8.89×10^{-2}	kg
	Chemicals organic	–	18.29	kg
	Natural gas	48.58	–	m^3
	Sulfuric acid	12.27	7.3×10^{-3}	t
	Hydrochloric acid	1.66	–	t
	Soft water	2.33	–	t
Emission to air	Carbon dioxide	398.19	331.49	kg
	Mercury	5.75×10^{-2}	–	g
	Particulates	167.09	39.32	g
	Sulfur dioxide	566.05	400.38	g
	Nitrogen oxides	–	946.84	g
	Arsenic	0.34	4.04	g
	Lead	–	0.22	g
	Copper	–	0.39	g
	Nickel	–	8.48×10^{-3}	g
	Sulfuric acid	–	0.21	g
Emission to water	Phosphorus	2.56×10^{-2}	1.54×10^{-2}	g
	COD	19.56	–	g
	Nitrogen	1.93	–	g
	Zinc	3.83×10^{-2}	–	g
	Copper	2.34×10^{-2}	0.19	g
	Sulfide	2.13×10^{-3}	–	g
	Lead	2.23×10^{-2}	1.65×10^{-2}	g
	Nickel	–	0.24	g
	Cadmium	1.28×10^{-2}	–	g
	Mercury	4.46×10^{-5}	–	g
	Ammonia	–	0.20	g
	Arsenic	1.68×10^{-2}	6.51×10^{-3}	g
	Fluoride	0.30	–	g
	Chromium VI	1.70×10^{-3}	–	g
	Petroleum oil	1.70×10^{-2}	–	g
Solid waste	Drilling waste	14.72	–	t
	MSW to landfill	–	1.06	kg
	Hazardous waste	–	4.66	kg

human toxicity and rest categories, respectively. For the reclaimed copper scenario, the variation of direct emissions was the highest for climate change and human toxicity in which a 5% reduction led to a decrease of 16.57 kg CO_2 eq and

Table 2 Element content rate of raw materials and wastes

	Copper (%)	Sulfur (%)	Arsenic (%)	Lead (%)	Zinc (%)
Copper concentrate	23.7	27.42	0.06	0.93	1.78
Crude copper	97.4	0.09	0.03	0.28	0.83
Refined copper	99.9	–	–	–	–
Coke	–	0.6	–	–	–
Diesel	–	0.22	–	–	–
Smelt smoke dust	5.2	7.71	0.71	33.63	6.78
Crude nickel sulfate	0.3	6.4	–	1.5	–
Anode mud	20	8.5	0.51	15	0.93
Gangue	0.2	0.08	0.02	0.86	2.77
Water quenched slag	0.5	0.85	0.05	0.4	2.71
Arsenic slag	2.9	19.3	24.5	0.46	0.36
Sodium sulfide	–	24.6	–	–	–
Sulfuric acid	–	32.65	–	–	–
Gypsum	–	12.6	–	–	–

10.54 kg 1,4-DB eq, respectively. The reduction of heavy fuel oil represented the largest variation in the metal and fossil depletion categories. By contrast, the impact of the variation of wastewater from the reclaimed copper production minimally influenced all impact categories. Similar results were obtained in soft water generated from the refined copper production.

3.3 Uncertainty analysis

Uncertainty analysis results, which were presented as geometric square deviation (GSD^2), were indicated for each scenario to estimate the degree of confidence for the midpoint results (see Table 3). For the climate change category, the midpoint values were 1.91×10^3 and 688.62 kg CO_2 eq, and the GSD^2 values were 1.24 and 1.21 for refined and reclaimed copper, respectively. These results mean that a 95% confidence interval corresponds to integration over the climate change score range from 1.54×10^3 to 2.39×10^3 kg CO_2 eq and 569.11 to 833.23 kg CO_2 eq for refined and reclaimed copper, respectively. The probability of refined copper or reclaimed copper was 100%, thereby indicating that the climate change score of the refined copper scenario is higher than that of the reclaimed copper scenario. For the remaining categories, a similar analogy could be made with GSD^2 and the probability results (see Table 5). The

Table 3 Life cycle midpoint results for refined and reclaimed copper production. Values are presented per function unit

Categories	Unit	Refined copper	Reclaimed copper
Climate change	kg CO_2 eq	1.91×10^3	688.62
Ozone depletion	kg CFC-11 eq	4.15×10^{-5}	3.66×10^{-5}
Terrestrial acidification	kg SO_2 eq	6.56	2.34
Freshwater eutrophication	kg P eq	1.73×10^{-2}	1.98×10^{-3}
Marine eutrophication	kg N eq	0.32	9.38×10^{-2}
Human toxicity	kg 1,4-DB eq	99.77	221.56
Photochemical oxidant formation	kg NMVOC	6.58	2.28
Particulate matter formation	kg PM_{10} eq	2.72	0.84
Terrestrial ecotoxicity	kg 1,4-DB eq	0.36	4.02×10^{-2}
Freshwater ecotoxicity	kg 1,4-DB eq	3.54	0.16
Marine ecotoxicity	kg 1,4-DB eq	3.28	0.32
Ionizing radiation	kq U235 eq	17.99	7.71
Agricultural land occupation	m^2a	23.29	0.66
Urban land occupation	m^2a	12.60	2.13
Natural land transformation	m^2	0.12	0.15
Water depletion	m^3	34.85	8.45
Metal depletion	kg Fe eq	2.36×10^4	4.63
Fossil depletion	kg oil eq	655.72	269.06

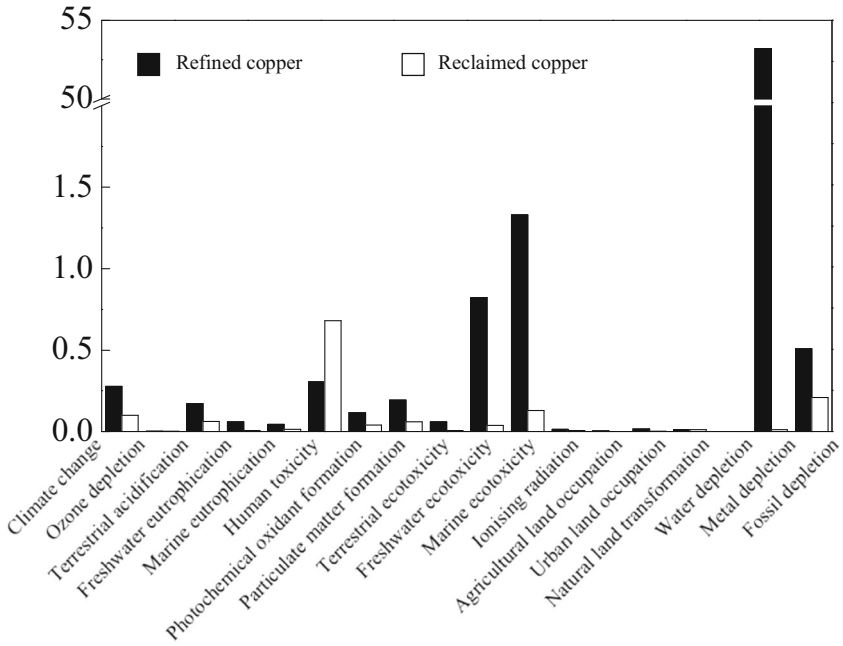


Fig. 2 Normalized LCIA midpoint scores for the full life cycle

environmental burden of the refined copper scenario was higher than that of the reclaimed copper scenario in most key categories except for human toxicity, in which a similar environmental burden between both scenarios was observed.

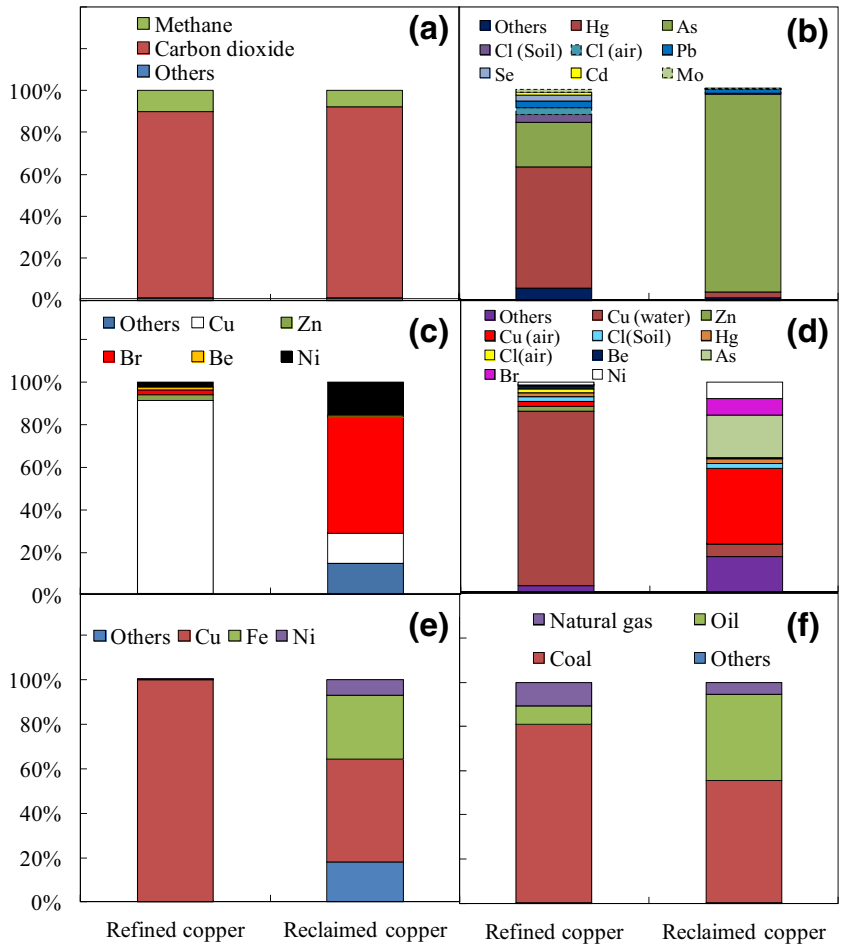


Fig. 3 Substance contribution. **a** Climate change. **b** Human toxicity. **c** Freshwater ecotoxicity. **d** Marine ecotoxicity. **e** Metal depletion. **f** Fossil depletion

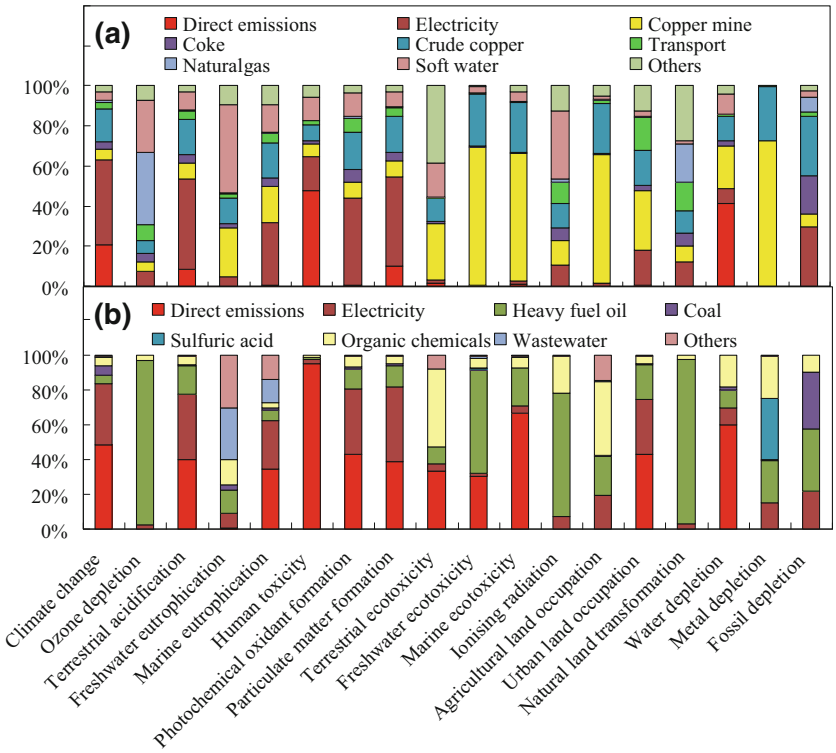


Fig. 4 Process contribution. **a** Electronic copper. **b** Reclaimed copper

3.4 Mass balance

The copper mass balance of refined scenario was presented to further understand the reliability of the LCI (see Table 6). The initial amount of copper was 1.04 and 1.01 tons for per ton copper output. This result indicated that the loss rate was 3.29% for the refined copper. The lost copper mass may be attributed to the limitations related to the missing inventory and measurement error problems. For the rest elements, a similar analogy can be made with the results shown in Table 6. Conclusively, more than 94% elements were involved, thereby indicating that the inventory presented in this study is within a reasonable range.

The aforementioned results indicate that the reclaimed copper scenario represented 59 to 99% environmental benefits in most key categories except for human toxicity, in which a similar environmental burden was observed (see Tables 4 and 5) because of the direct atmospheric arsenic emission during copper recycling stage. The increasing copper recycling rate and improvement of the air controlling system for arsenic in the reclaimed copper industry are the key factors in reducing the overall environmental burden generated from

Table 4 Sensitivity of main contributors. Values are presented per functional unit

	Processes	Climate change (kg CO_2 eq)	Human toxicity (kg 1,4-DB eq)	Freshwater ecotoxicity (kg 1,4-DB eq)	Marine ecotoxicity (kg 1,4-DB eq)	Metal depletion (kg Fe eq)	Fossil depletion (kg oil eq)
	Variation	5%	5%	5%	5%	5%	5%
Refined copper	Direct emissions	19.91	2.37	0	0	0	0
	Electricity consumption	40.57	0.85	0	0	0.12	9.75
	Copper mine	4.77	0.32	0.12	0.10	857.52	2.07
	Crude copper	15.81	0.40	0.05	0.04	323.66	9.63
	Soft water	4.29	0.59	0.01	0.01	0.11	1.11
Reclaimed copper	Direct emissions	16.57	10.54	0	0.01	0	0
	Electricity consumption	12.16	0.26	0	0	0.04	2.92
	Heavy fuel oil	1.72	0.15	0	0	0.06	4.78
	Organic chemicals	1.78	0.07	0	0	0.06	1.29
	Wastewater	0.06	0.01	0	0	0	0.01

Table 5 Uncertainty analysis results presented by geometric square deviation (GSD^2) of each scenario in copper production

	$GSD^2_{\text{refined copper}}$	$GSD^2_{\text{reclaimed copper}}$	$\text{Probability}_{\text{refined copper} \geq \text{reclaimed copper}}$
Climate change	1.24	1.21	100%
Human toxicity	1.64	4.33	22.5%
Freshwater ecotoxicity	4.07	2.18	100%
Marine ecotoxicity	3.41	1.95	100%
Metal depletion	1.40	1.39	100%
Fossil depletion	1.20	1.20	100%

the copper industry in China. In 2013, copper recycling rate (22.8%, China Nonferrous Metals Industry Association 2014) was higher than that reported in Japan (16.7%; World Metal Statistics Yearbook 2013), USA (3.8%; World Metal Statistics Yearbook 2013), and Russia (14.9%; World Metal Statistics Yearbook 2013) but lower than that reported in Germany (29.4%, World Metal Statistics Yearbook 2013). If 1% copper recycling rate and arsenic controlling rate are improved, approximately 0.3, 1.4, and 0.3% environmental impacts in the climate change, metal depletion, and fossil depletion categories can be reduced. For the refinery and reclaimed copper scenarios, approximately 0.2 and 0.9% human toxicity burden can be reduced, respectively. In addition, the arsenic concentrations emitted from the refined copper smelting stage presented in this study (0.34 g/t) are significantly lower than those reported by the European database (2.0–32.5 g/t; Process ID: EI3CQUNI 00006401520237 4; EI3ARUNI 00011519603682; Ecoinvent center 2015). By contrast, the arsenic concentrations presented in the reclaimed copper scenario (4.0 g/t) are significantly higher than those shown in the European database (3.5×10^{-3}–2.0 g/t; Process ID: Standard 000034454913626; Standard 000034454913627; Ecoinvent center 2015). This result indicated that the improvement potential for atmospheric arsenic control in the reclaimed copper industry in China was high. To improve the air quality in China, the Chinese government implemented a new limit on arsenic emission for the reclaimed copper plants on July 1, 2015 (GB 31574-2015). The required upper limit of arsenic and its compounds is 0.4 mg/m^3, which is lower than the concentration (0.14–3.0 mg/m^3) presented in this study. If the arsenic emission from the reclaimed copper scenario is the same with that from the refined scenario, then approximately 75% environmental benefit can be achieved in the human toxicity category.

Song et al. (2014) reported an LCA case study of the refined and reclaimed copper production in China, which

Table 6 Mass balance of refined copper production process (unit: t)

		Amount	Copper	Sulfur	Arsenic	Lead	Zinc
Input	Copper concentrate	3.26	0.77	0.89	1.96×10^{-3}	3.03×10^{-2}	5.80×10^{-2}
	Crude copper	0.28	0.27	2.52×10^{-4}	8.40×10^{-5}	7.84×10^{-4}	2.32×10^{-3}
	Sodium sulfide	1.61×10^{-3}	–	3.95×10^{-4}	–	–	–
	Coke	1.57×10^{-1}	–	9.45×10^{-4}	–	–	–
	Diesel	1.00×10^{-3}	–	2.21×10^{-6}	–	–	–
	Total		1.04	8.95×10^{-1}	2.04×10^{-3}	3.11×10^{-2}	6.03×10^{-2}
Output	Refined copper	1.00	1.00	–	–	–	–
	Smelt smoke dust	5.36×10^{-2}	2.79×10^{-3}	4.13×10^{-3}	3.80×10^{-4}	1.80×10^{-2}	3.63×10^{-3}
	Sulfuric acid	2.53	–	0.82	–	–	–
	Crude nickel sulfate	1.43×10^{-2}	4.30×10^{-5}	9.18×10^{-4}	–	2.15×10^{-4}	–
	Anode mud	7.95×10^{-3}	1.59×10^{-3}	6.76×10^{-4}	4.05×10^{-5}	1.19×10^{-3}	7.39×10^{-5}
	Gangue	5.19×10^{-1}	1.04×10^{-3}	4.15×10^{-4}	1.04×10^{-4}	4.46×10^{-3}	1.44×10^{-2}
	Water quenched slag	1.47	7.34×10^{-3}	1.25×10^{-2}	7.34×10^{-4}	5.87×10^{-3}	3.98×10^{-2}
	Arsenic slag	2.72×10^{-3}	7.88×10^{-5}	5.24×10^{-4}	6.65×10^{-4}	1.25×10^{-5}	9.78×10^{-6}
	Gypsum	9.49×10^{-2}	–	1.20×10^{-2}	–	–	*
	Direct air emission		–	2.83×10^{-4}	3.40×10^{-7}	–	–
	Direct water emission		2.34×10^{-8}	2.13×10^{-11}	1.68×10^{-8}	2.23×10^{-8}	3.83×10^{-8}
	Total		1.01	8.56×10^{-1}	1.92×10^{-3}	2.98×10^{-2}	5.79×10^{-2}
Loss rate			3.29%	4.56%	5.96%	4.44%	4.23%

employed the Eco-Indicator 99 method based on the background inventory data shown in the GaBi database. The results indicated that the environmental impacts for refined copper production were mainly generated from the coal mining stage, which is consistent with the results presented in the current study (Figs. 2 and 4 and Table 4). Memary et al. (2012) developed a time-series LCA for copper mining and smelting in Australia since the 1940. Results showed that global warming potential for per ton copper production ranged from approximately 2.5 to 8.5 t-CO_2 eq. Similarly, Reuter et al. (2015) reported that the global warming potential score for copper production in Europe was ranged from approximately 2.6 to 5.6 t-CO_2 eq/t-copper, which was higher than the result (1.9 t-CO_2 eq/t-copper) presented in the current study. However, the values in the primary energy demand and global warming impact for the refined copper production were approximately 1.5 to 2 times higher than those in European ecoinvent database (i.e., Process ID: EI3CQUNI 000064015202374; EI3ARUNI 00011519603682; Ecoinvent center 2015) because of the varying of energy structure between Europe and China, the overuse of energy in China, and the missing inventory of carbon dioxide emissions in the European database. Coal-based electricity generation is used in this study because coal plays an important role in China's energy structure (National Bureau of Statistic of China 2015). If the current hydropower share rate (17%, National Bureau of Statistic of China 2015) is considered in this study, approximately 7 and 14% of the primary energy demand and global warming impacts, respectively, can be reduced. Thus, copper production with clean energy, such as hydropower, is crucial to reduce the overall environmental burden. However, copper products are currently produced in Shandong, Henan, Jiangsu, Anhui, and Jiangxi with a relatively low hydroelectricity share rate (China Nonferrous Metals Industry Yearbook 2015; China Energy Statistical Yearbook 2014). Thus, the relocation of the copper industry near areas with abundant clean energy sources is highly recommended.

4 Conclusions

This research evaluated the environmental impact and the improvement potentials of the copper production in China by conducting LCA. Uncertainty analysis is also performed to increase the reliability of this study. The main conclusions drawn from the present study are:

- The emissions of carbon dioxide, mercury, arsenic, and copper, use of coal and oil, and mining copper ore were the most significant contributors to the overall environmental burden.
- The direct emission level of arsenic generated by the reclaimed copper industry in China is significantly higher than that of the Europe technologies.
- Compared with the refined copper scenario, approximately 59 to 99% environmental benefits in each key category will be achieved for the reclaimed copper scenario.
- Efficient methods to reduce the overall environmental impact of copper production are increasing the direct heavy metal emissions in air and water, the recycling rate of copper, electricity consumption efficiency, the utilization rate of clean energy sources, and the efficiency of copper ore mining and consumption.

The results presented in this study can potentially expand the life cycle inventory database of copper production in China, as well as provide useful information for policy-makers in China in making decisions regarding the eco-design, management, education, policy making, and problem rectification of the copper industry. However, other technologies (e.g., flash smelting, flash converting, oxygen bottom/side blowing melting, Baiyin copper smelting, copper hydrometallurgy, and blast furnace smelting) should be continued to be studied. Additionally, the economic and social impacts on the copper industries in China should be further addressed for more sustainable development. Further research on this area will be conducted by our research group.

Acknowledgments We gratefully acknowledge financial support from the Thirteen-Five national key research and development program (Grant nos. 2017YFF0206702;2017YFF0211605), National Natural Science Foundation of China (Grant nos. 71671105), and the Institute of the Fundamental Research Funds of Shandong University (2015JC016).

References

Ayer N, Martin S, Dwyer R, Gace L, Laurin L (2016) Environmental performance of copper-alloy net-pens: life cycle assessment of Atlantic salmon grow-out in copper-alloy and nylon net-pens. Aquaculture 453:93–103

Chen W, Hong J, Xu C (2015) Pollutants generated by cement production in China, their impacts, and the potential for environmental improvement. J Clean Prod 103:61–69

Chen W, Zhang F, Hong J, Shi W, Feng S, Tan X, Geng Y (2016) Life cycle toxicity assessment on deep-brine well drilling. J Clean Prod 112:326–332

China Energy Statistical Yearbook (2014) Department of Energy Statistics, National Bureau of Statistics. China Statistics Press

China Nonferrous Metals Industry Yearbook (2008–2015) China Nonferrous Metals Industry Yearbook Editorial Board

China Nonferrous Metals Industry Association (2014) China Nonferrous Metals Industry Yearbook in 2013. China Nonferrous Metals Industry Association, Beijing

China Nonferrous Metals Industry Association (2015) China Nonferrous Metals Industry Yearbook in 2014. China Nonferrous Metals Industry Association, Beijing

China Statistical Yearbook (2005-2015) China statistics press, Beijing. Available from: <http://www.stats.gov.cn/tjsj/ndsj/>

Ecoinvent centre (2015) Swiss centre for life cycle inventories. Available from: <http://www.ecoinvent.org/database/>

GB 31574-2015. Emission standards of pollutants for secondary copper, aluminum, lead and zinc industry. Ministry of Environmental Protection of the people's Republic of China and General Administration of Quality Supervision, Inspection and Quarantine of the People's Republic of China

Giurco D, Stewart M, Suljada T, Petrie J (2006) Copper recycling alternatives: an environmental analysis. Proceedings of the 5th annual environmental engineering research event, Noosa, Australia: 20-23

Goedkoop M, Heijungs R, Huijbregts M, De Schryver A, Struijs J, Van Zelm R A (2009) Life cycle impact assessment method which comprises harmonised category indicators at the mid-point and the end-point level. ReCiPe, first edition report I: characterization. Available from: <http://www.lciarecipe.net>

Goedkoop M, Spriensma R (2001) The eco-indicator99: a damage oriented method for life cycle impact assessment. Methodology report: 1-144

Guinée J B, Gorrée M, Heijungs R, Huppes G, Kleijin R, de Koning A, van Oers L, Wegener Sleeswijk A, Suh S, Udo de Haes H.A, de Bruijn H, van Duin R, Huijbregts M.A.J, Lindeijer E, Roorda A.A.H, Weidema B.P (2001) Life cycle assessment: an operational guide to the ISO standards, part 3: scientific background. Ministry of Housing, Spatial Planning and Environment (VROM) and Centre of Environmental Science (CML), Den Hag and Leiden, the Netherlands

Hong J, Chen Y, Wang M, Ye L, Qi C, Yuan H, Zheng T, Li X (2017a) Intensification of municipal solid waste disposal in China. Renew Sust Energ Rev 69:168–176

Hong J, Han X, Chen Y, Ye L, Qi C, Li X (2017b) Life cycle environmental assessment of industrial hazardous waste incineration and landfilling in China. Int J Life Cycle Assess 22(7):1054–1064

HJ 559-2010 (2010) Environmental protection standard of the people's Republic of China. Cleaner production standard, Copper electricity industry. Ministry of Environmental Protection of the people's Republic of China

ISO (2006a) Environmental management-life cycle assessment-general principles and framework. ISO 14040, Geneva, Switzerland

ISO (2006b) Environmental management-life cycle assessment requirements and guidelines. ISO 14044 Geneva, Switzerland

Li Y, Guan J (2009) Life cycle assessment of recycling copper process from copper-slag. Int Energ Environ Technol 1:198–201

Memary R, Giurco D, Mudd G, Mason L (2012) Life cycle assessment: a time-series analysis of copper. J Clean Prod 33:97–108

National Bureau of Statistic of China (2015) China Statistical Yearbook 2014. Available from: <http://www.stats.gov.cn/tjsj/ndsj/2014/indexch.htm>. Accessed 19 Oct 2017

Reuter MA, Schaik A, Gediga J (2015) Simulation-based design for resource efficiency of metal production and recycling systems: cases-copper production and recycling, eWaste (LED lamps), nickel pig iron. Int J Life Cycle Assess 20:671–693

Rubin R, de Castro M, Brandão D, Schalch V, Ometto A (2014) Utilization of life cycle assessment methodology to compare two strategies for recovery of copper from printed circuit board scrap. J Clean Prod 64:297–305

Schryver A, Brakkee K, Goedkoop M, Huijbregts M (2009) Characterization factors for global warming in life cycle assessment based on damages to humans and ecosystems. Environ Sci Technol 43:1689–1695

Streicher-Porte M, Althaus H (2010) China and global markets: copper supply chain sustainable development: a life cycle assessment study. Materials science and technology, technology and society laboratory Lerchenfeldstrasse 5 CH-9014 St. Gallen, Switzerland. Available from: <https://www.iisd.org/sites/default/files/publications/china_copper_supply.pdf>

Song X, Yang J, Lu B, Li B, Zeng G (2014) Identification and assessment of environmental burdens of Chinese copper production from a life cycle perspective. Front Env Sci Eng 8(4):580–588

Wang H, Liu Y, Gong X, Wang Z, Gao F, Nie Z (2015) Life cycle assessment of metallic copper produced by the pyrometallurgical technology of China. Mater Sci Forum 814:559–563

Wegener Sleeswijk A, Van Oers L, Guinée J, Struijs J, Huijbregts M (2008) Normalisation in product life cycle assessment: an LCA of the global and European economic systems in the year 2000. Sci Total Environ 390:227–240

World Metal Statistics Yearbook (2013) World Bureau of Metal Statistics. Available from: <http://www.world-bureau.com/index.html>

Xu C, Shi W, Hong J, Zhang F, Chen W (2015) Life cycle assessment of food waste-based biogas generation. Renew Sust Energ Rev 49:169–177

Zhao T, Gong S, Zhang X, Jaffe D (2008) Asian dust storm influence on North American ambient PM levels: observational evidence and controlling factors. Atmos Chem Phys 8:2717–2728